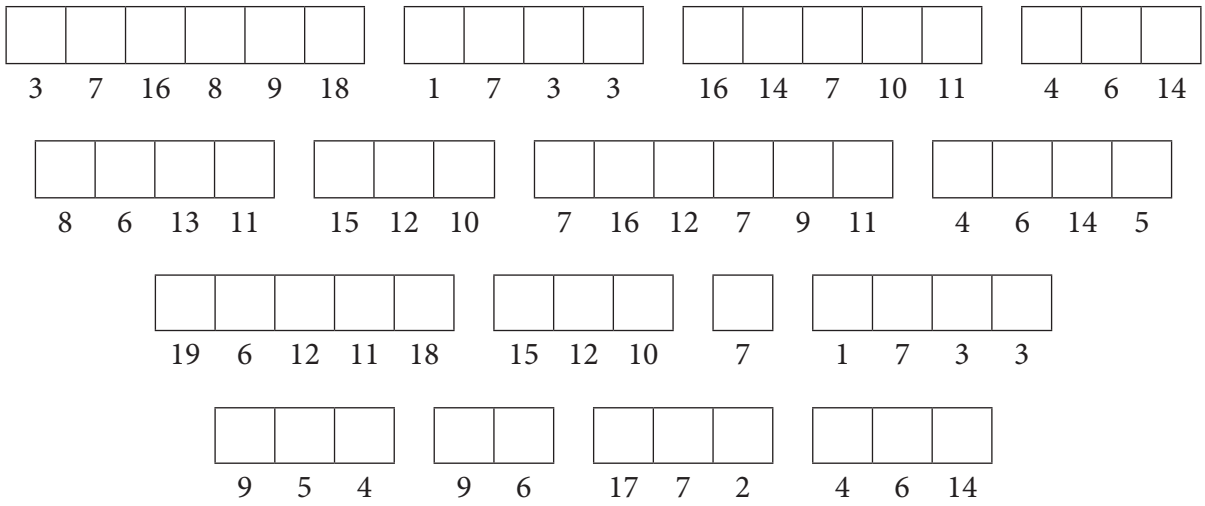
WORKSHEET

Chapter 7: Integral calculus



Coldplay (Fix You)

Questions

Find the following integrals.

1 $\int \frac{dx}{1+x^2}$

2 $\int \frac{dx}{4+x^2}$

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3 $\int \frac{dx}{1+9x^2}$

4 $\int \frac{dx}{\sqrt{1-4x^2}}$

5 $\int \frac{8x}{\sqrt{1-4x^2}}dx$

6 $\int \frac{\sin(x)}{\sqrt{1+2\cos(x)}}dx$

7 $\int \frac{e^{2x}}{e^{2x}-3}dx$

8 $\int (4x-1)^7\,dx$

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9 $\int \frac{-3}{(3x-7)^2} dx$

10 $\int \frac{dx}{2x^2+4x+3}$

11 $\int \frac{x}{\sqrt{x+3}} dx$

12 $\int x^2 e^{x^3}\, dx$

13 Write $\frac{1}{9-x^2}$ in the partial fraction form $\frac{A}{3-x} + \frac{B}{3+x}$.

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14 Using partial fractions, find the integral $\int \frac{dx}{9-x^2}$.

15 First find the derivative of the expression $x\sin(x)$ and hence find the integral $\int x\cos(x)dx$.

Evaluate the following definite integrals.

16 $\int_0^{\frac{\pi}{2}} \cos(x)dx$

17 $\int_0^{\frac{\pi}{4}} \sin(x)\cos^2(x)dx$

18 $\int_0^{\frac{\pi}{4}} \frac{\sin(x)}{2\cos(x)}dx$

19 $\int_0^2 \frac{3x^2}{x^3-2}dx$

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Solutions

A $x\sin(x) + \cos(x) + c$

B $\ln(3)$

D $\frac{1}{\sqrt{2}}\arctan\left(\sqrt{2}x + \sqrt{2}\right) + c$

E $\frac{2(x+3)^{\frac{3}{2}}}{3} - 6(x+3)^{\frac{1}{2}} + c$

F $\frac{-1}{3}\left(\frac{\sqrt{2}}{4} - 1\right)$

G 1

H $\frac{(4x-1)^8}{32} + c$

I $\frac{1}{2}\ln\left|e^{2x} - 3\right| + c$

L $\frac{1}{3}\arctan(3x) + c$

M $\frac{1}{6}\left(\frac{1}{3-x} + \frac{1}{3+x}\right)$

N $\frac{e^{x^3}}{3} + c$

O $-\sqrt{1 + 2\cos(x)} + c$

R $-2\sqrt{1 - 4x^2} + c$

S $\frac{-1}{2}\ln\left(\frac{1}{\sqrt{2}}\right)$

T $\frac{1}{3x-7} + c$

U $\frac{1}{6}\ln\left|\frac{3+x}{3-x}\right| + c$

W $\arctan(x) + c$

X $\frac{1}{2}\arctan\left(\frac{x}{2}\right) + c$

Y $\frac{1}{2}\arcsin(2x) + c$

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